



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Exam Name: Mid Term Exam Trimester: Spring 2026

Course Code: PMG 4101, Course Title: Project Management

Total Marks: 30

Duration: 1 hour 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions.

Consider the following Scenario:

The government of Bangladesh has assigned your team to develop a Digital Family Card–Based Smart Welfare Distribution Platform to support low-income and vulnerable families across the country. The main goal is to make sure financial aid and subsidies reach the right people in a fair and transparent way. The system will create a centralized platform where household data is stored digitally, connected with national ID systems, and shared across government offices and financial service providers to improve coordination.

The platform will include features like digital family registration, AI-based eligibility checking, and automatic monthly payments through mobile financial services and banks. Each family will get a smart Family Card with QR or NFC, linked to national ID data to avoid duplication and fraud. Government officials will be able to monitor fund distribution, track beneficiaries, and analyze poverty trends using dashboards. Local offices will handle registration and verification, while citizens can use mobile apps or SMS to check payments, get updates, and manage their information.

The system needs to work with existing systems like national ID databases and financial platforms. It also has to handle challenges like large amounts of data, limited internet in rural areas, and keeping information accurate. The interface should be simple and easy to use, especially for people who are not very familiar with technology, and it should work even with low internet or offline when needed.

Security is very important since the system will store sensitive personal and financial data, so strong authentication, encryption, and monitoring will be used. AI will help in decision-making by predicting poverty trends, detecting unusual activities, and improving how resources are distributed. The system will be developed step by step, starting with basic features like registration and payments, and later adding advanced features like analytics, fraud detection, and full integration with other systems.

To complete this project successfully, collaboration will be needed between government agencies, project managers, developers, AI experts, financial institutions, and local authorities. The aim is to build a system that is reliable, scalable, and actually helpful for people across Bangladesh.

1. Choose the most appropriate **Software Engineering Methodology** for the project scenario and explain why your chosen methodology is superior for this project. Define the concept of **Product Backlog** and **Daily Scrum** according to SCRUM methodology

[CO1]

3+2

2. Discuss '**Project Plan**' in software project planning and state its **elements**. Write down the **Vision** and **Scope** Document for the project described above following proper outline.

[CO2] 1+4

3. As a project manager if you want to generate a proper Work Breakdown Structure (WBS), state the rules that you would suggest to your team. Develop the **WBS** for the scenario described above and **estimate** (duration unit of month) for **two major WBS pieces/components** for the above scenario following the principles and rules.

[CO3] 1+3+1

4. Write down activities of the **moderator** during the '**Kick off Meeting**' for Delphi Wideband Estimation (DWE) technique. Suppose you want to form an estimation team consisting of **five members**. Show the steps of estimation session (**one estimation form** and **tentative convergence chart** of duration) for the above-described project, according to DWE method.

[CO3] 1+4

5. Give a brief explanation of the **RMMM** concept. Now as a project manager you are preparing for Risk Mitigation. Develop a **Risk Plan Sheet** showing at least **three** major risks for the above scenario.

[CO2] 2+3

6. Progress report submission and presentation on Work Breakdown Structure and Delphi Wideband Estimation Technique for a group project have been evaluated earlier (**Do not answer this question, already completed in classroom**).

[CO3] 5